

Aliah University

End-Semester Examination (Autumn Semester) - 2026

(For 2nd Year 3rd Semester MCA)

Paper Name: Information Security

Full Marks: 80

Paper Code: MCAPGPC18

Time: 3 hrs

Group-A

(Answer all questions)

(5 × 2 = 10)

1.

- Define Cryptanalysis. [CO1, BL2]
- Explain Honeypots. [CO3, BL2]
- Explain Transpositional cipher with example. [CO2, BL2]
- Convert the given text "ALGORITHM" into cipher text using monoalphabetic substitution with key=4. [CO4, BL4]
- Define : Cybersecurity. [CO1, BL1]

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Group-B

(Answer any five questions)

(5×6= 30)

2. Explain various types of active attacks. (6) [CO3, BL2]
3. Given the prime numbers $p=11$ and $q=17$, try to find out N , e , d using RSA. (6) [CO4, BL4]
4. Explain Caesar and Vigenere cipher using examples. (3+3=6) [CO3, BL2]
5. Explain DMZ network in details with diagram. (6) [CO5, BL4]
6. Explain block cipher and stream cipher with examples. (2+4=6) [CO1, BL1]
7. What are Tactics, Techniques, and Procedures (TTPs) in Cyber security? (6) [CO2, BL3]

Group-C

(Answer any four questions)

(4 × 10 = 40)

8. Explain the algorithm for generating keys in RSA algorithm. Perform encryption and decryption using RSA algorithm. for the following. $P=7$; $q=11$; $e=13$; $M=8$. (3+7=10) [CO2, BL5]
9. What is threat? Explain man in the middle attack with diagram. (2+8=10) [CO2, BL4]
10. Illustrate Digital Signature with neat diagram.. (2+8=10) [CO2, BL3]
11. Discuss ESP in detail with its packet format. (10) [CO5, BL2]
12. What is Honeynet? Explain Intrusion Detection System in details. (2+8=10) [CO4, BL3]

RSA